DSCUC Assignment – Short Answers

B.Tech CSE (AI) · Semester III

Q1. Explain asymptotic notations.

Big O (O) gives an asymptotic upper bound; **Big Omega (Ω)** gives a lower bound; and **Big Theta (Θ)** gives a tight bound. For example, a loop that runs n times has time complexity O(n).

Q2. Explain arrays and their basic operations.

An array stores elements of the same data type in contiguous memory locations, accessed by index. Basic operations are traversal, insertion, deletion, searching, and updating. Index access takes O(1); traversal and general insertion/deletion take O(n).

Q3. What is a sparse matrix? Explain its representation.

A sparse matrix contains mostly zero values. It can be stored efficiently using a three-tuple representation: each entry stores the row index, column index, and non-zero value. The first row stores the number of rows, columns, and non-zero elements.

Example matrix:

```
[[0, 0, 5], [0, 8, 0], [3, 0, 0]]
```

Three-tuple form (row, column, value): (3, 3, 3), (0, 2, 5), (1, 1, 8), (2, 0, 3).

Q4. What is recursion? Explain with an example.

Recursion is a technique in which a function calls itself. It needs a base case to stop and a recursive case to reduce the problem. For factorial: n! = 1 when n = 0; otherwise n! = n × (n−1)!.

Example: 5! = 5 × 4 × 3 × 2 × 1 = 120.

Q5. Convert infix to postfix using a stack.

Infix places operators between operands (A+B); postfix places operators after operands (AB+). Scan left to right: output operands, push operators onto a stack according to precedence, handle parentheses, then pop remaining operators.

```
Example: A+B*(C−D) → ABCD−*+
```